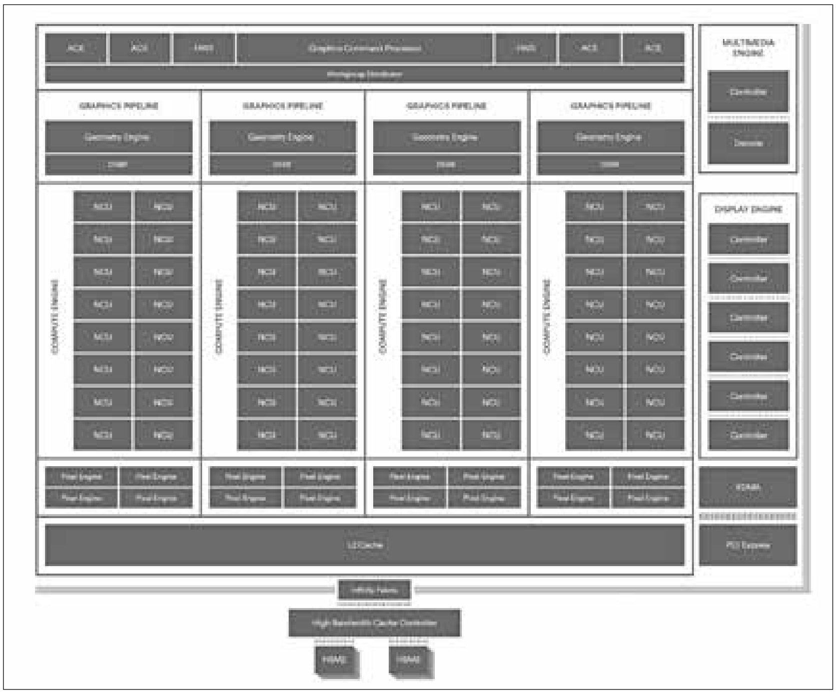
hardware for these operations for some added acceleration.

AMD

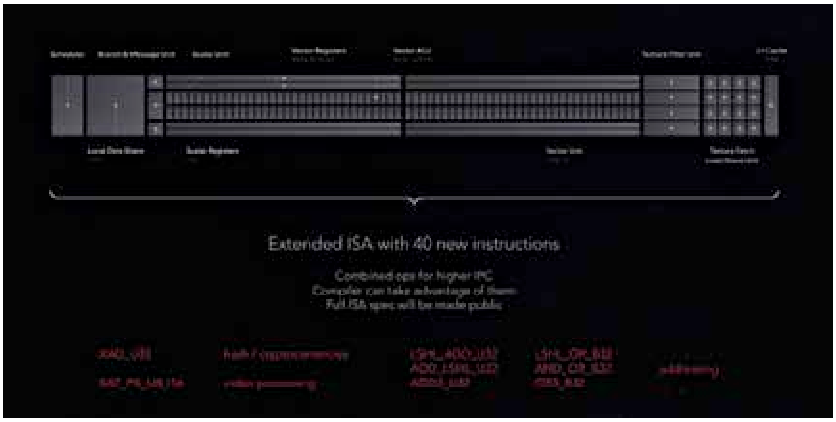
AMD's graphics architectures are called Graphics Core Next or GCN for short. As of the latest GPU release, the Radeon VII, they're on GCN 5.0 with the upcoming NAVI GPUs to have GCN 6.0 or whatever they'll be calling it then. To understand the microarchitecture of GCN 5.0, let's take a look at the Radeon Vega 64. It's got 4096 Stream Processors which are grouped into subdivisions called Compute Units (CU). In all, the Vega64 has 64 CUs.

There's a High Bandwidth Cache sub-system which allows more memory than what the graphics card has. The GPU will be able to use your system memory as an extension to its own in this case. The 64 NCUs are further grouped together into 4 shader engines or as AMD calls them, Asynchronous Compute Engines. Each Compute Engine gets one NGG engine and a rasteriser along with 16 ROPs.

With GCN 5.0, the CUs were re-vamped by AMD and are now being called Next Generation Compute Units (NCU). Each NCU consists of 64 SIMD units, 4 Texture Units, 64 KB of shared cache and 16 KB of L1 cache. Additionally, there's a Next Generation Geometry (NGG) engine



Vega64 GCN 5.0 Block Diagram



New Generation Compute Unit breakdown

COMPARING GRAPHICS PERFORMANCE

In a similar fashion as CPUs, even the GPUs can be compared against each other based on the workloads. With graphics cards, there exists the same issue as CPUs i.e. proprietary APIs. Both NVIDIA and AMD have their own set of APIs for enhanced effects such as shadows, simulating hair, etc. If a game developer chooses to implement these APIs, then that particular game will perform better on the respective company's GPUs. This also means that performance might be adversely affected on GPUs which don't support these APIs. Take for example the infamous Crysis tessallation usaage. Tessallation had just been introduced into mainstream AAA titles back then and Crysis went overboard with absolutely unnecessary tessallation layers. Imagine this, there was an ocean under which there existed an invisible tessallation layer. You could never see it but it used up a lot of compute resources of the graphics card. NVIDIA's cards had much improved tessallation performance and with these unnecessary invisible layers, it seemed as if the developers were wantonly making the game run slower on AMD GPUs. AMD is the good guy here. Most of their graphics APIs are open-source so developers could implement them and improve the visual fidelity in games. NVIDIA followed suit in 2016 and made their Gameworks APIs open source as well.

Coming back to the comparison aspect, you really need to figure out which APIs are being used to know which graphics cards will perform better. Certain studios have tie-ups with GPU manufacturers for all their games so if you're a fan of that particular studio, then you might want to rethink your GPU choice. At the end, with gaming, it's all about the FPS values and how they scale at different resolutions. Certain graphics cards will excel at low resolutions because the GPU is powerful but as you scale the resolution, the VRAM becomes a limiting factor and you'll see frame drops across the board. Ideally, you want to focus on a graphics card with more unified shaders and a high memory pool. With each passing year, the amount of memory consumed by video games has only risen, so if you want to experience better looking textures go for a card with more memory. To summarise, your priority should be Unified shader count > Memory > Graphics APIs.